

Module 1: Web & Internet Foundations

- Understanding how the internet works
- Client–server communication model
- Version Control (Git & Github)
- What are AI tools, LLMs, and Copilots (high-level overview)
- Conceptual understanding of tools like ChatGPT, Cursor, and Antigravity
- Prompting fundamentals for developers
- Ethical and responsible use of intelligent development tools

Cursor / Antigravity – Developer Workflow Integration

- Conceptual explanations of request–response flow
- Visual breakdown of how browsers and servers interact

Module 2: HTML Structure & Semantics

- Structuring data using tables
- Block vs inline elements
- Building forms with various input types
- HTML5 attributes and best practices
- Writing semantic and meaningful HTML
- Introduction to accessibility and ARIA

Module 3: CSS Fundamentals

- Applying styles using inline, internal, and external CSS
- Selectors and specificity
- Pseudo-classes and pseudo-elements
- Colors and measurement units
- Fonts, typography, and text styling

Module 4: CSS Layout & Responsive Design

- CSS box model
- Positioning techniques
- Flexbox layout system
- CSS Grid for complex layouts
- Responsive design with media queries

Module 5: UI Frameworks (Bootstrap & Tailwind)

- Typography systems
- Responsive breakpoints
- Grid-based layouts
- Forms and buttons
- Navigation bars and reusable UI components

Module 6: JavaScript Runtime & Environment

- JavaScript language overview
- Browser vs Node.js environments
- Execution context and call stack
- Memory management basics
- Debugging with browser developer tools
- Installing and using Node.js and npm

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- Runtime error analysis using Cursor
- Visual understanding of call stack and execution flow

Module 7: JavaScript Core Programming

- Variables and data types
- Operators and comparisons
- Scope and lexical environment
- Conditional logic
- Loops and iteration

Module 8: Advanced JavaScript Concepts

- Objects and object methods
- `this` keyword behavior
- Function binding: `call`, `apply`, `bind`
- Prototypal inheritance

- Rest and spread operators
- Shallow vs deep copy
- Closures and data encapsulation

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- Inline explanations for complex JS concepts
- Concept reinforcement using guided examples

Module 9: Modern JavaScript (ES6+)

- Callbacks and asynchronous basics
- Higher-order functions
- Function currying
- Array methods: map, filter, reduce
- IIFE and scope isolation
- JavaScript event loop

Module 10: DOM Manipulation

- DOM structure and traversal
- Event handling
- Event bubbling and delegation
- Interactive DOM-based mini projects

Module 11: Asynchronous JavaScript

- Callback hell and limitations
- Promises and chaining

- Fetch API
- async / await

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- Tracing asynchronous execution order
- Refactoring asynchronous code patterns

Module 12: React Fundamentals

- Virtual DOM
- React setup using Vite
- Components and JSX
- Props and data flow
- State management and event handling

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- Component structure validation
- Reviewing component responsibilities

Module 13: React Hooks – Core

- useState
- useEffect
- useRef
- Controlled and uncontrolled components

Module 14: Advanced React & Optimization

- useContext for global data sharing
- Custom hooks
- useMemo and useCallback
- React.memo

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- Identifying performance bottlenecks
- Optimization guidance for React applications

Module 15: React Routing

- Client-side routing with React Router
- Nested routes
- Dynamic route parameters
- Route protection and guards

Module 16: React Performance & UX

- Skeleton screens
- Lazy loading
- Suspense
- Shimmer loading effects

Module 17: State Management

- Global vs local state concepts
- Redux Toolkit fundamentals

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- Validating state ownership decisions
- Understanding Redux data flow

Module 18: Node.js Fundamentals

- Node.js architecture
- Event loop
- Module systems
- File system operations

Module 19: Express.js Basics

- HTTP fundamentals
- Express application setup
- Routing
- Request and response handling

Module 20: Express Middleware & Architecture

- CRUD API design
- Middleware execution flow

- Custom middleware
- Centralized error handling
- MVC architecture

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- Reviewing middleware sequencing
- Debugging request lifecycle issues

Module 21: REST API Development

- REST principles
- Scalable API design
- HTTP status codes
- Request validation
- API testing with Postman

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- API test case generation
- REST compliance review

Module 22: MongoDB & Mongoose

- NoSQL concepts
- MongoDB local and Atlas setup
- CRUD operations
- Schema and model design

Module 23: Authentication & Security

- Authentication and authorization flows
- Password hashing
- JWT-based authentication
- Role-based access control
- CORS and environment variables

Module 24: Capstone Project – Advanced Project Implementation

- System and architecture planning
- Backend API development
- Authentication and RBAC
- Frontend–backend integration
- Unit and API testing using Jest
- CI/CD automation
- Production deployment
- High-level understanding of RAG-based data retrieval

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- Daily development assistance using Cursor
- Test case generation
- System architecture and API flow visualization using Antigravity
- Exposure to real-world RAG usage patterns